Radxa E52C

Schematic

V1.2

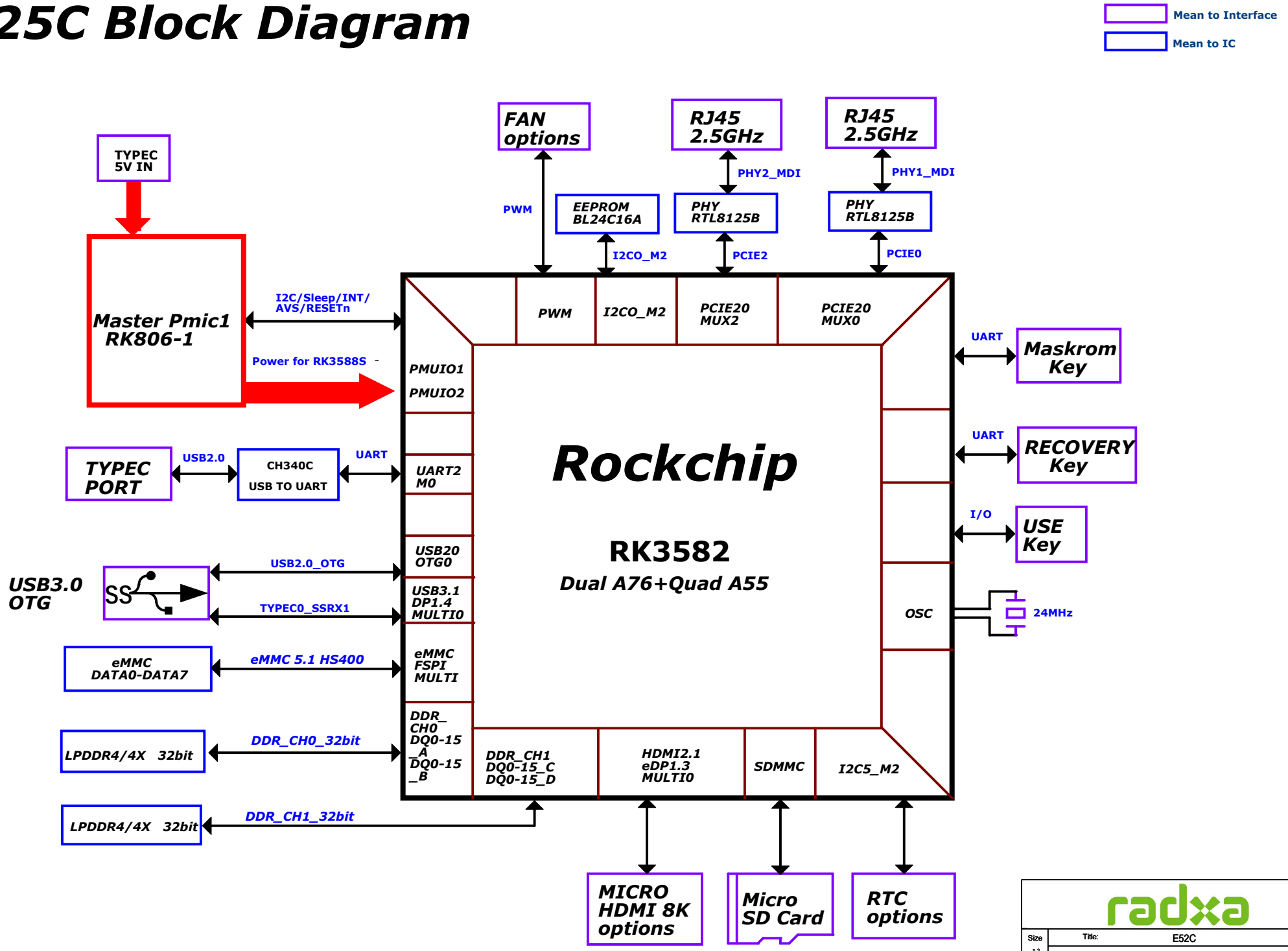
Table of Content

01.cover
02.Index
03.Block Diagram
04.RK3582_Power/GND
05.RK3582_OSC/PLL/PMUIO
06.RK3582_DDR Controler
07.RK3582 Flash/SD/USB
08.RK3582_SARADC/1.8V GPIO
09.RK3582_MIPI Interface
10.RK3582_PCIE/EDP
11.RK3582 GPIO
12.DRAM-LPDDR4/4X_2X32bit
13.MICRO HDMI 8K
14.UART/USB3 Port
15.TF Card/RTC/EEPROM
16.KEY/LED/FAN
17.EMMC FLASH
18.PCIE TO RJ45_1
19.PCIE TO RJ45_2
20.Power Ext
21.Power RK806-1
22.Power DC IN

radxa

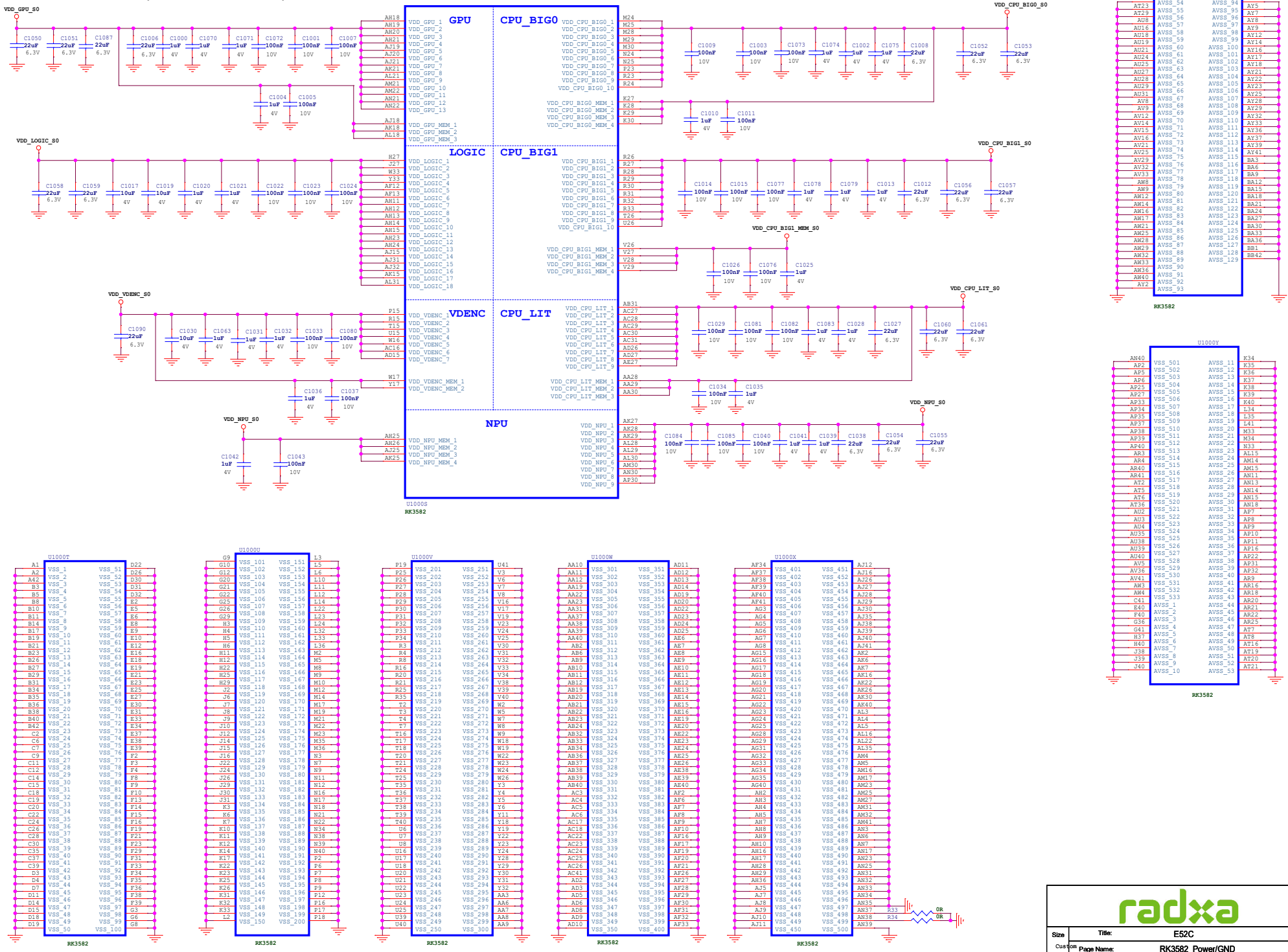
Size	Title:	REV
A4	E52C	V1.2
	Page Name: Index	
Date: Thursday, October 10, 2024	Sheet 02 of 22	

E25C Block Diagram



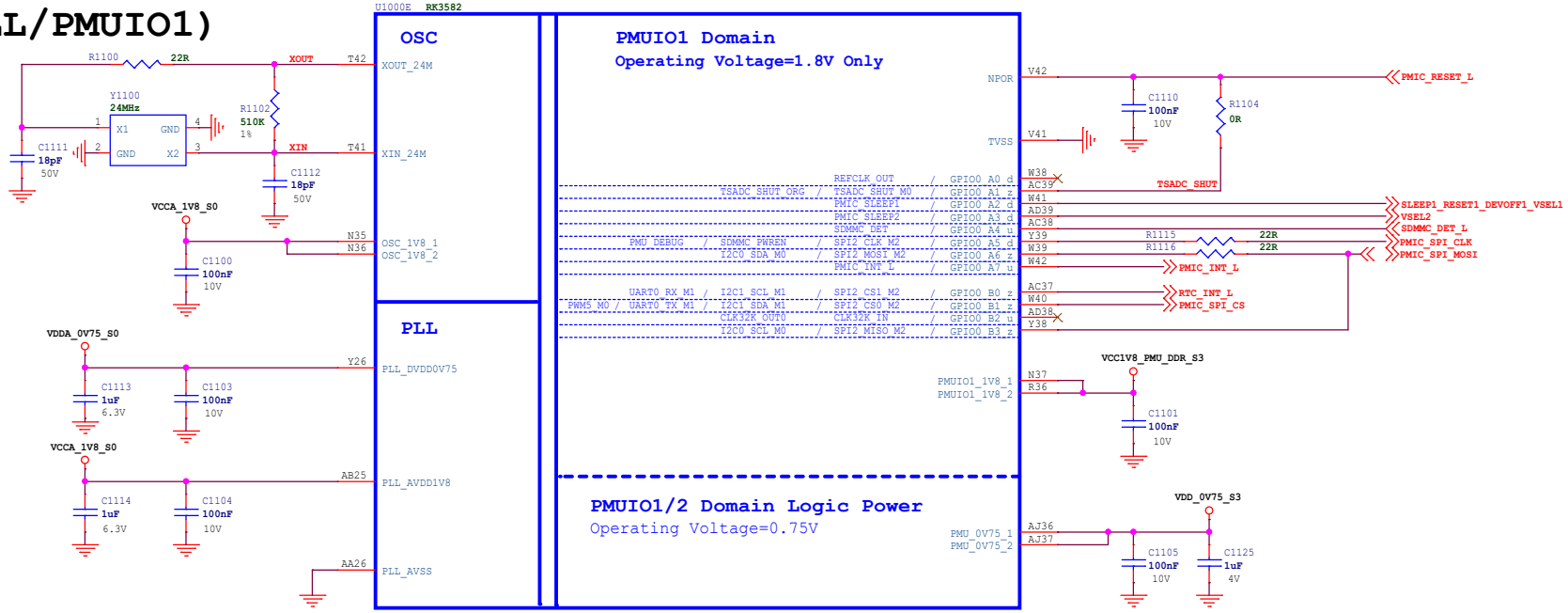
Size: A3	Title: E52C	REV: V1.2
	Page Name: Block Diagram	
Date: Thursday, October 10, 2024	Sheet 03 of 22	

RK3582 (Power&Gnd)

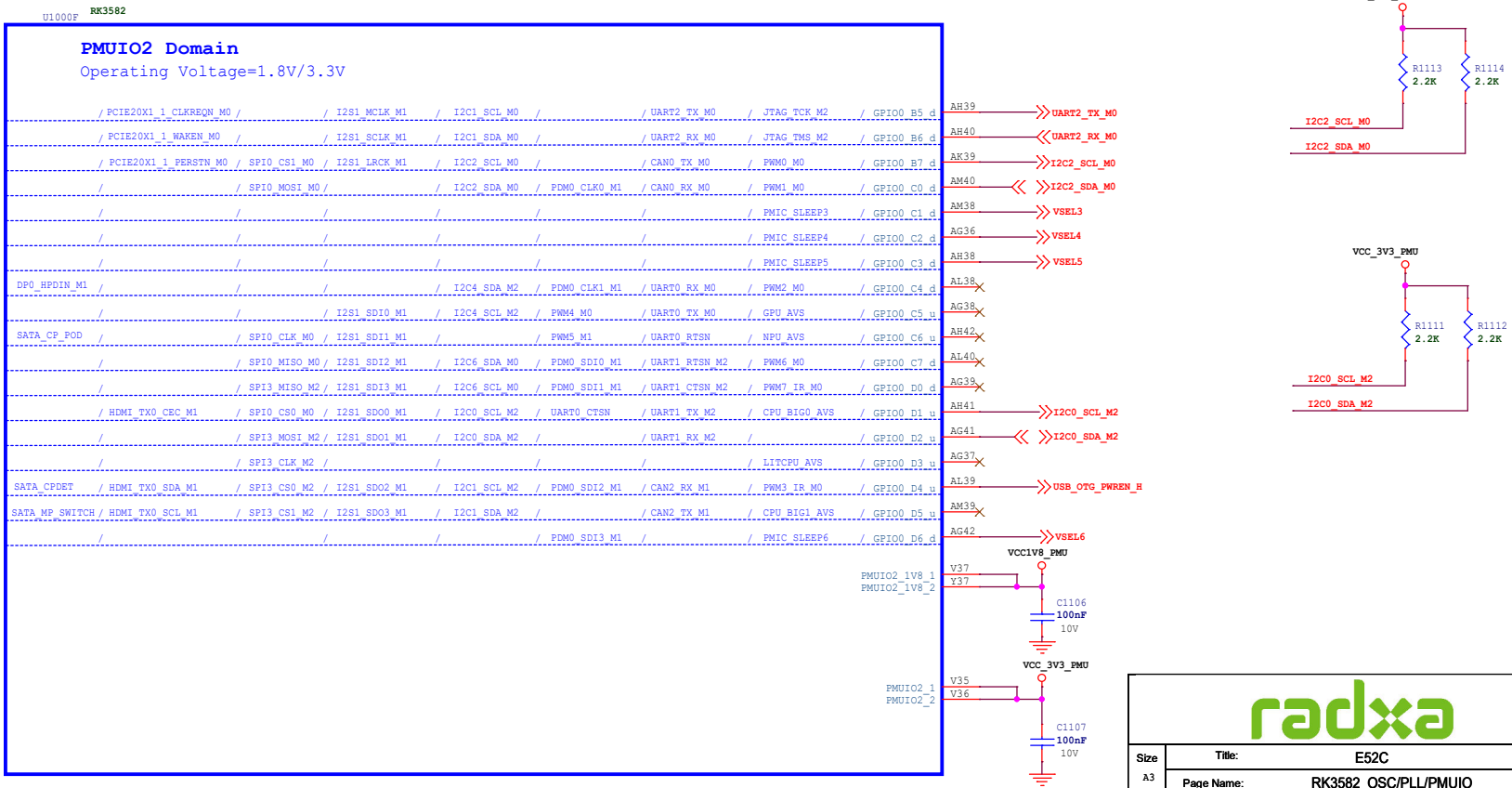


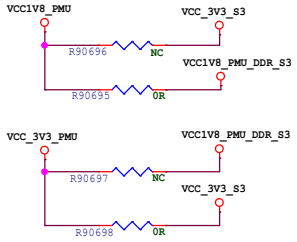
Size: Custom	Title: E52C		REV: V1.2
	Page Name: RK3582_Power/GND		
Date: Thursday, October 10, 2024	Sheet 04 of 22		

RK3582 (OSC/PLL/PMUIO1)



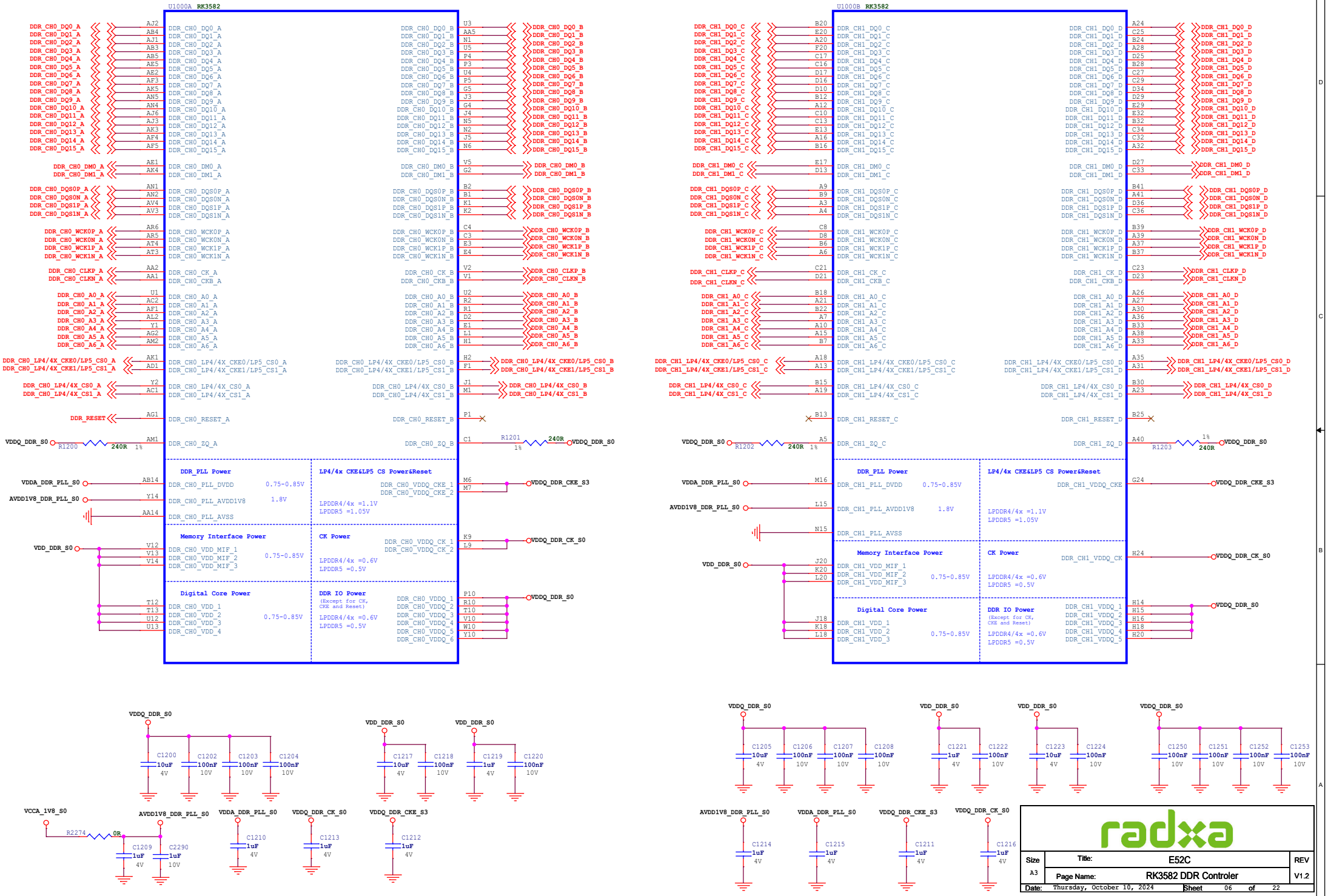
RK3582 (PMUIO2)



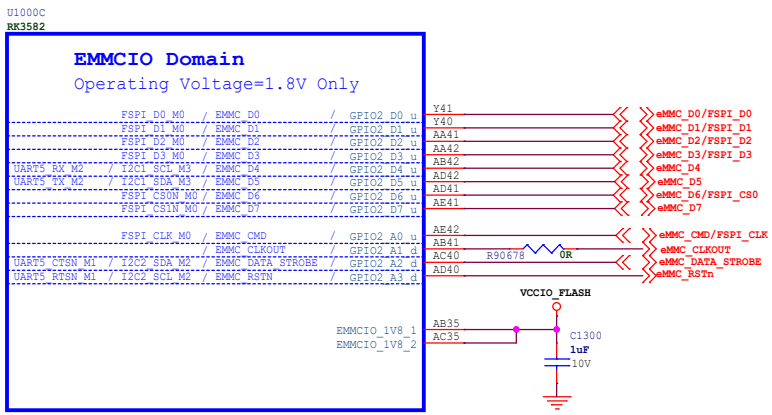


Size: A3	Title: E52C	REV: V1.2
	Page Name: RK3582_OSC/PLL/PMUIO	
Date: Thursday, October 10, 2024	Sheet 05 of 22	

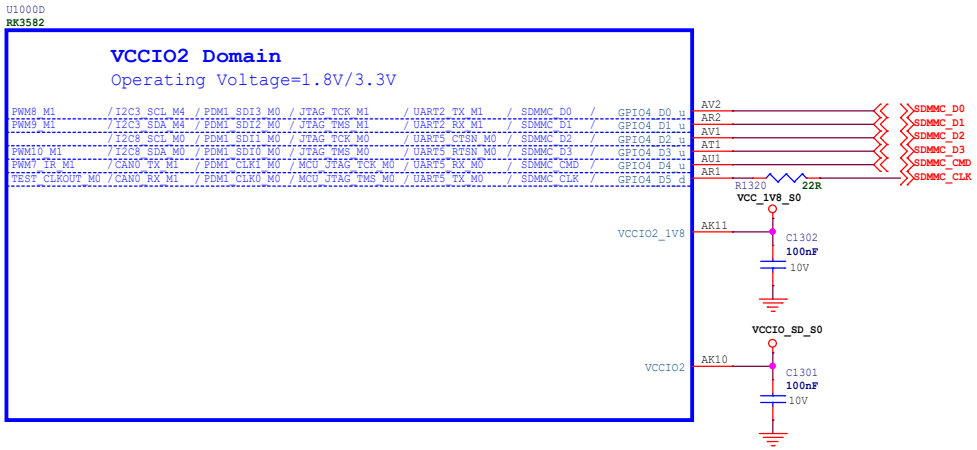
RK3582 (DDR PHY)



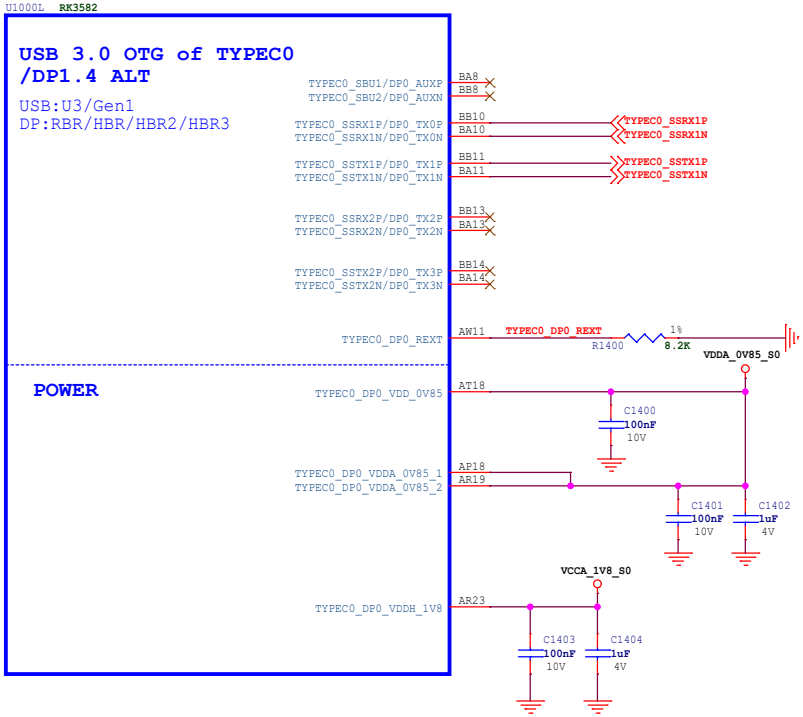
RK3582 (EMMCIO Domain)



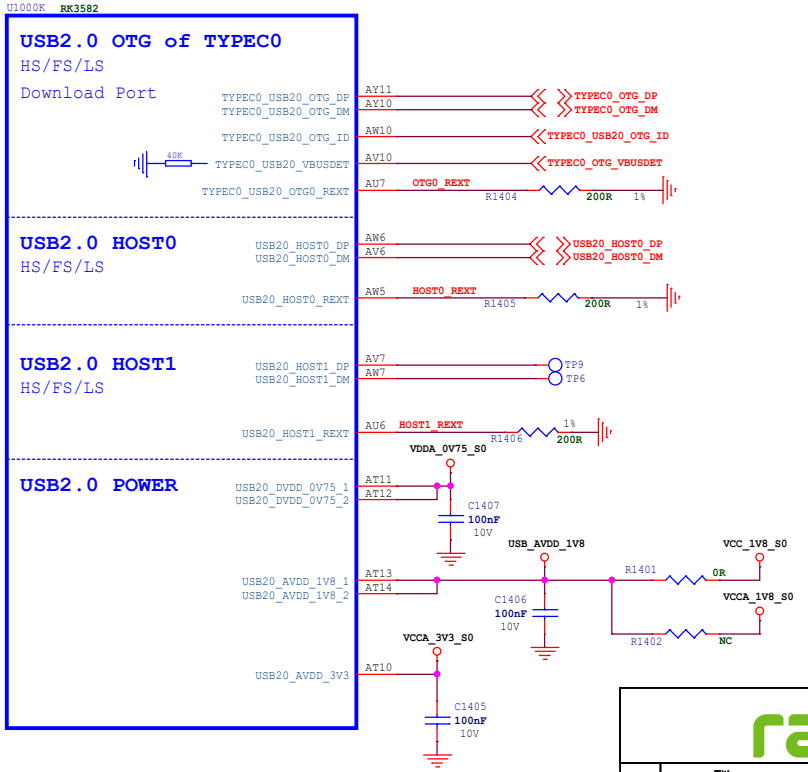
RK3582 (VCCIO2 Domain)



RK3582 (USB3.0/DP1.4)



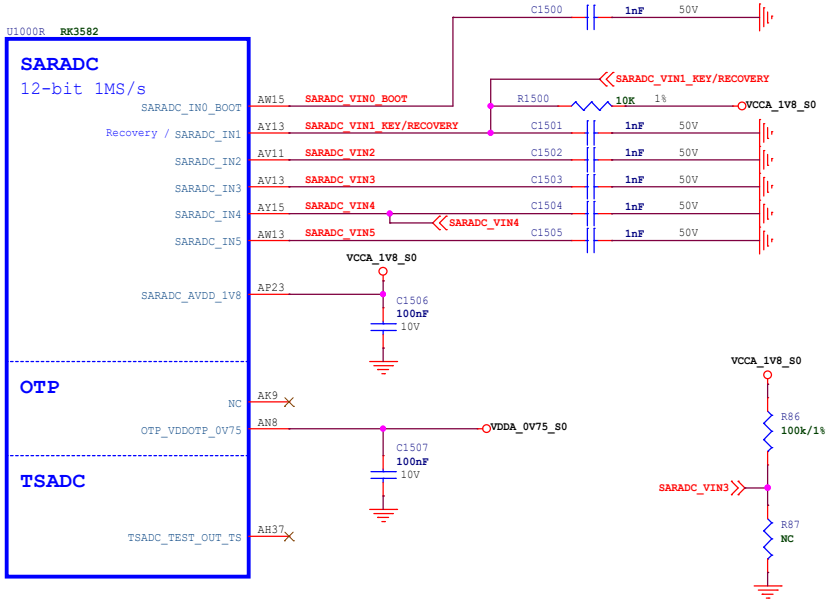
RK3582 (USB2.0)



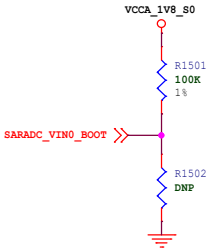
radxa

Size	Title:	REV
A3	E52C	V1.2
	Page Name: RK3582 Flash/SD/USB	
Date: Thursday, October 10, 2024	Sheet 07 of 22	

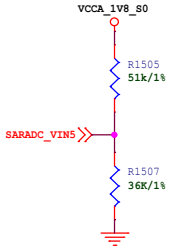
RK3582 (SARADC/OTP/TSADC)



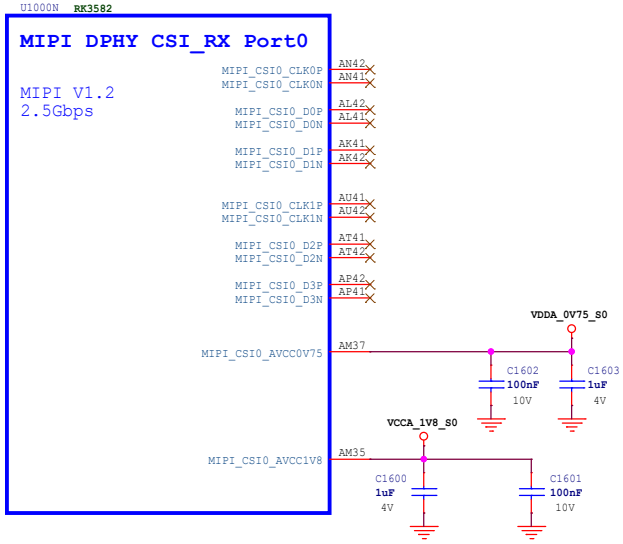
BOOT MODE CONFIG



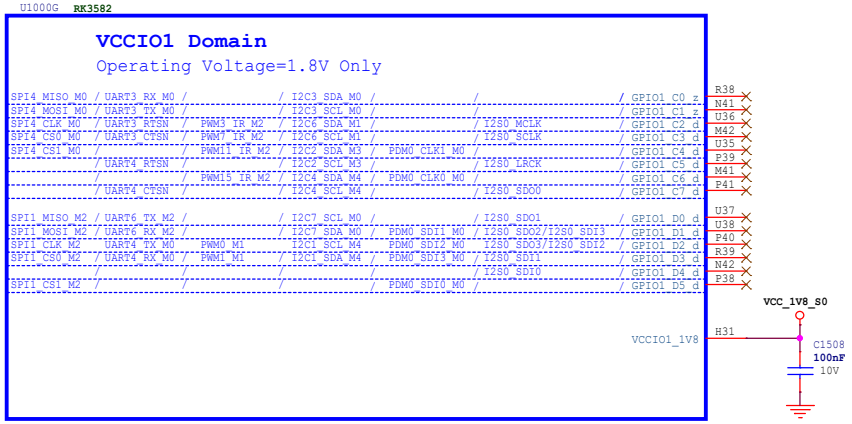
BOARD ID CONFIG



RK3582 (MIPI_DPHY CSI0 RX)

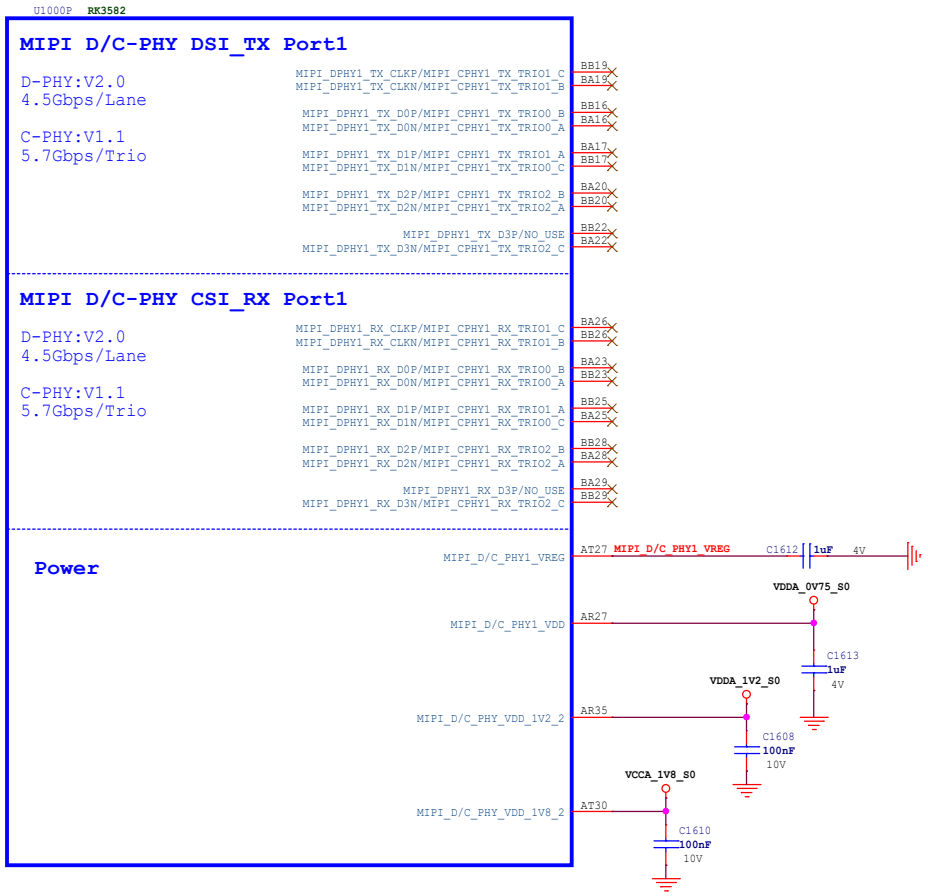


RK3582 (VCCIO1 Domain)

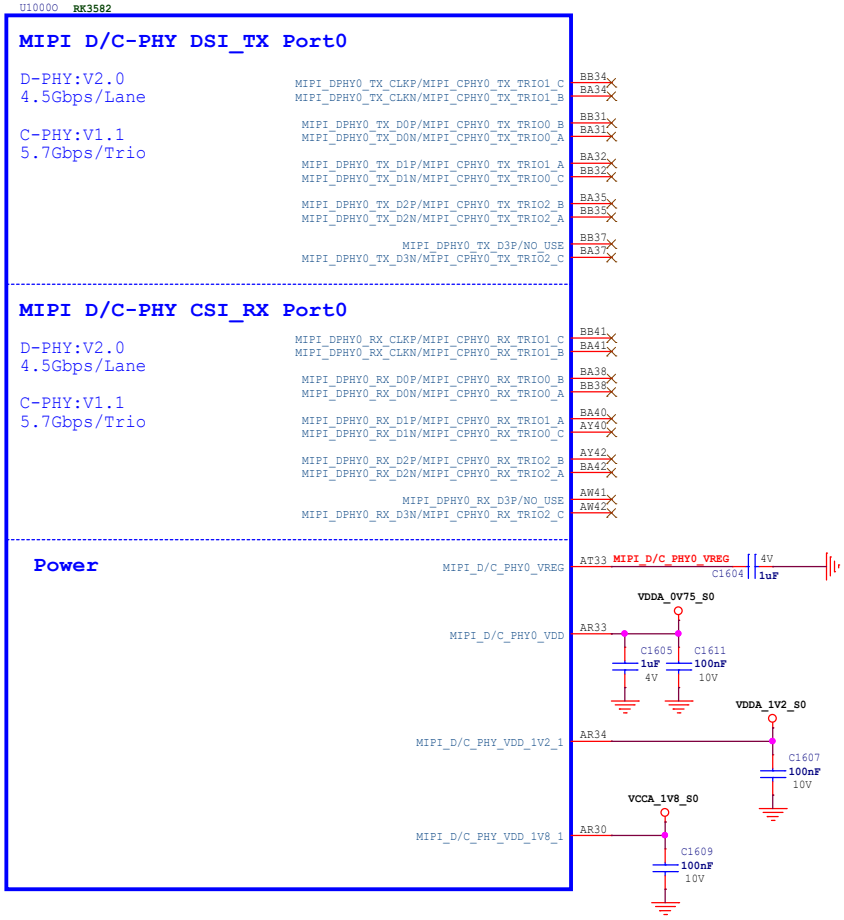


Size A3	Title: E52C		REV V1.2
	Page Name: RK3582_SARADC/1.8V GPIO		
Date: Thursday, October 10, 2024	Sheet 08 of 22		

RK3582(MIPI_D/C PHY1)

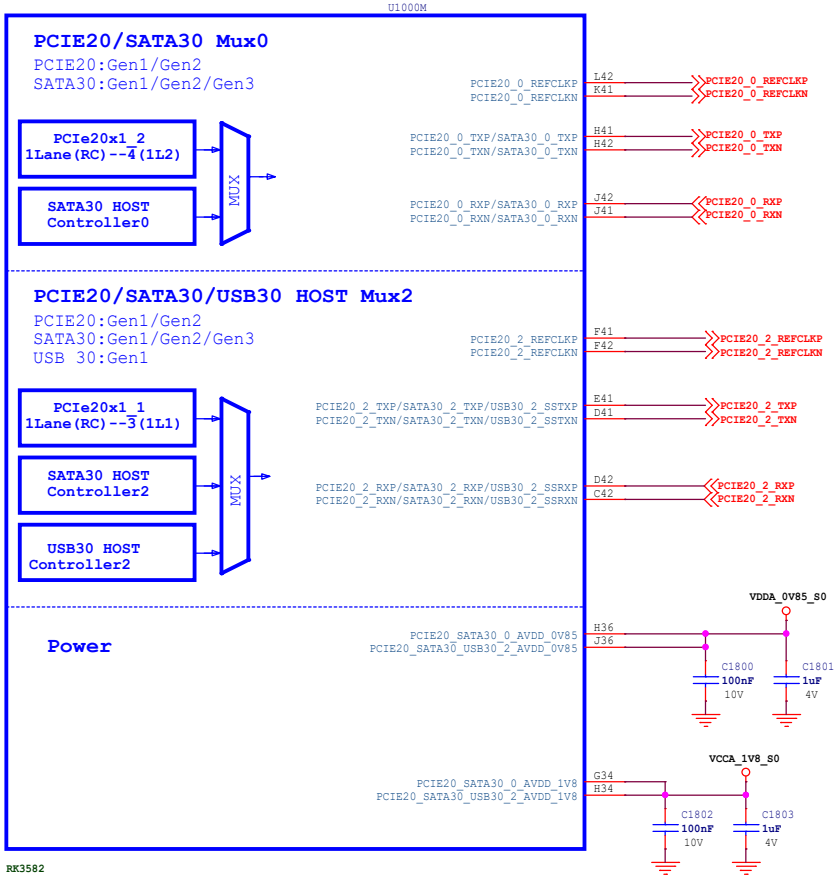


RK3582(MIPI_D/C PHY0)

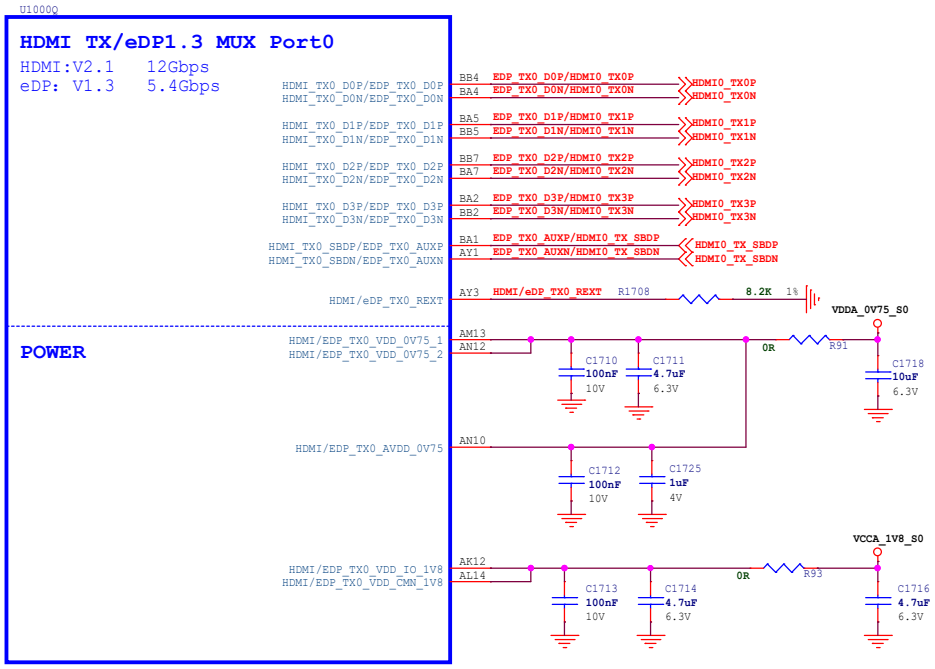


Size A3	Title: E52C	REV V1.2
	Page Name: RK3582_MIPI Interface	
Date: Thursday, October 10, 2024	Sheet 09 of 22	

RK3582(PCIE20/SATA30/USB30)

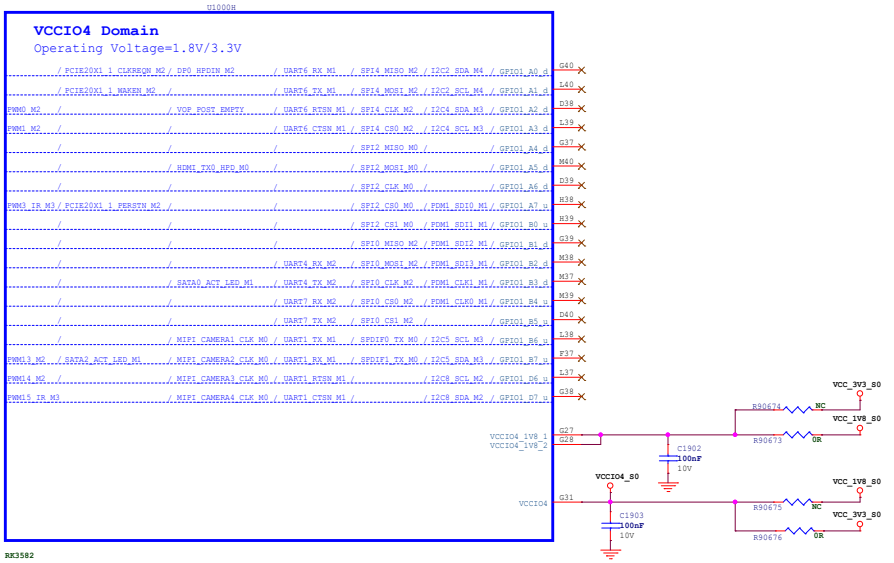


RK3582(HDMI2.1 TX/eDP1.3 TX)

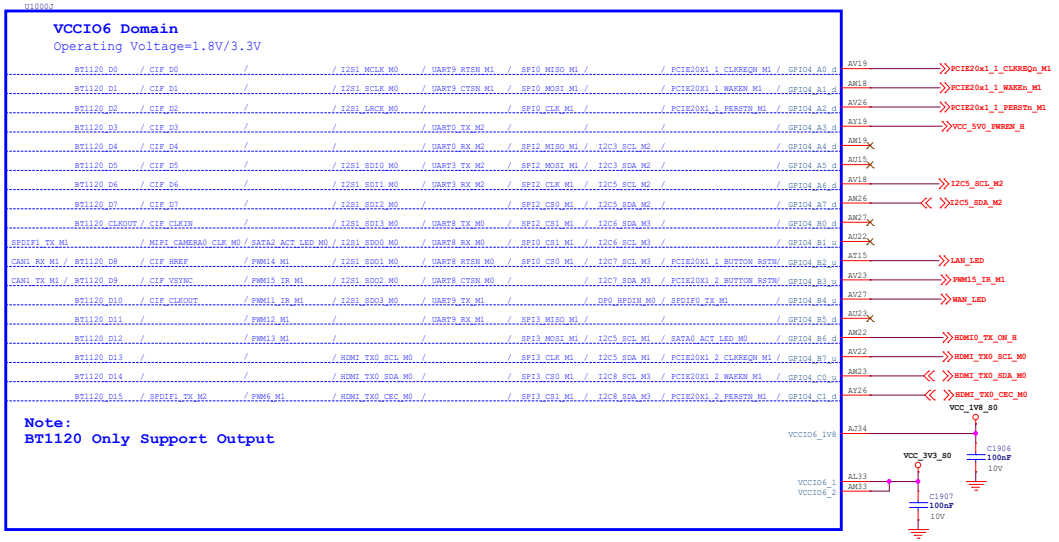


Size	Title:	REV
A3	E52C	V1.2
	Page Name: RK3582 PCIE/EDP	
Date: Thursday, October 10, 2024	Sheet 10 of 22	

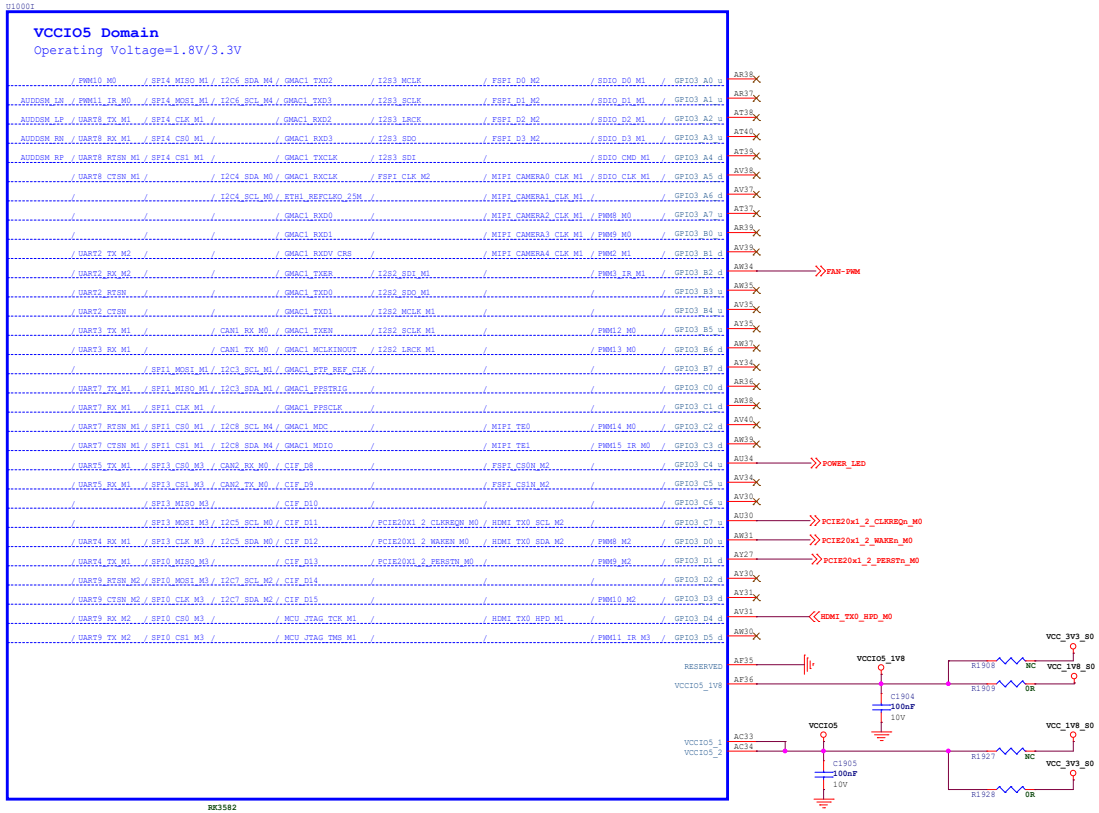
RK3582 (VCCIO4 Domain)



RK3582 (VCCIO6 Domain)

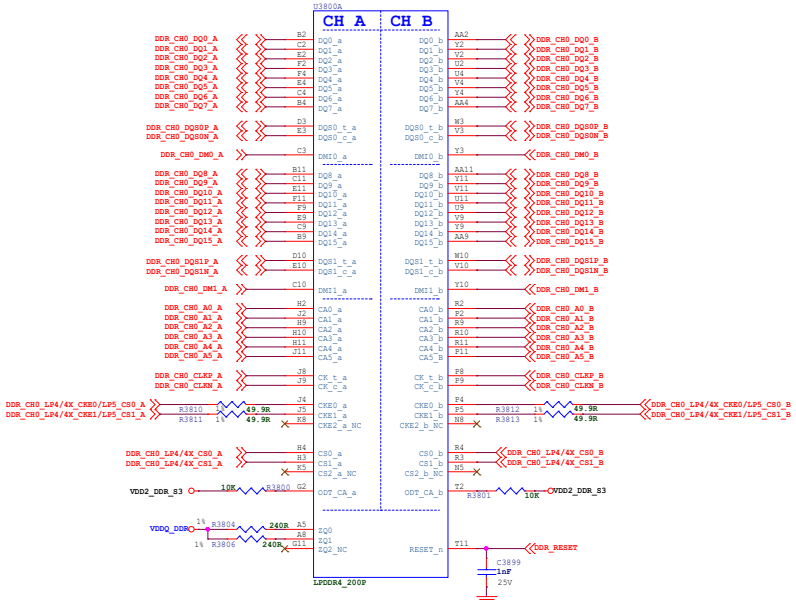


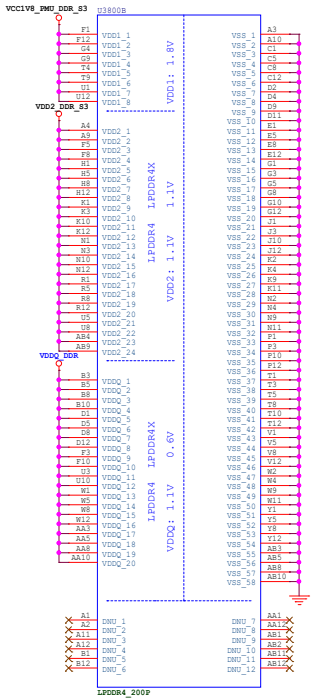
RK3582 (VCCIO5 Domain)

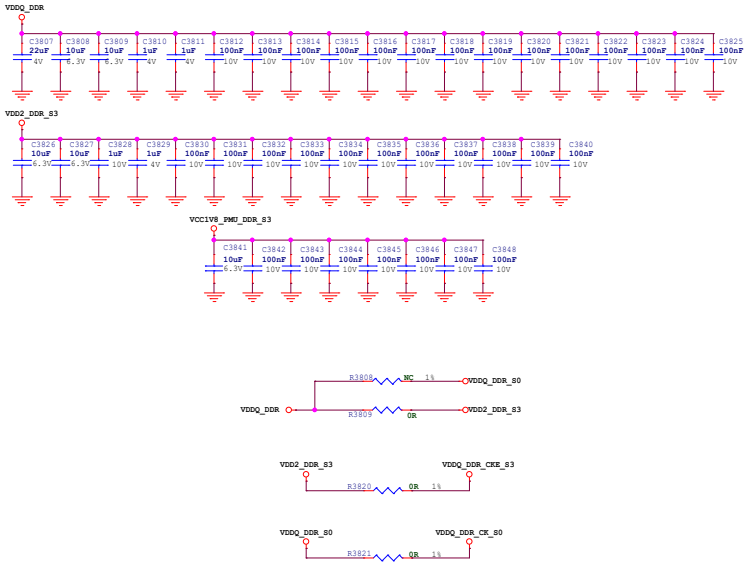


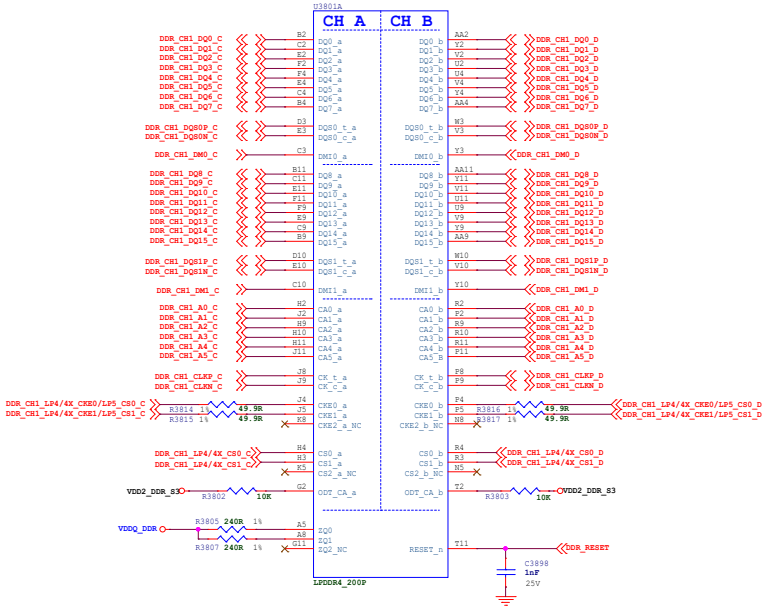
Size: A2	Title: E52C	REV: V1.2
	Page Name: RK3582 GPIO	
Date: Thursday, October 10, 2024	Sheet 11 of 22	

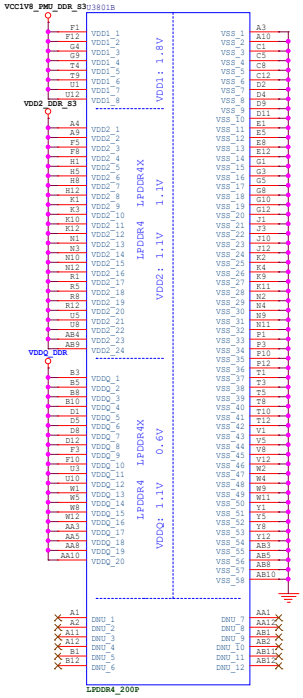
DRAM-LPDDR4/4x_2X32bit

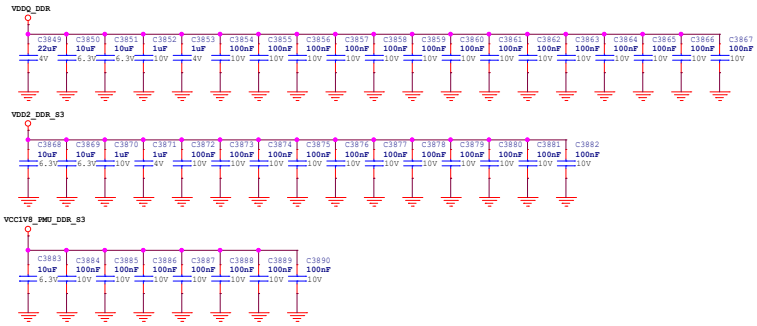






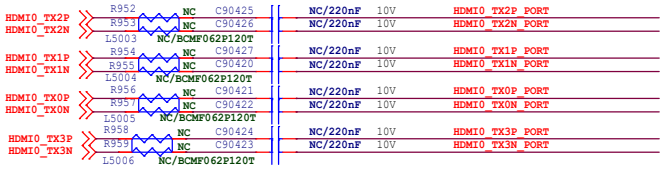


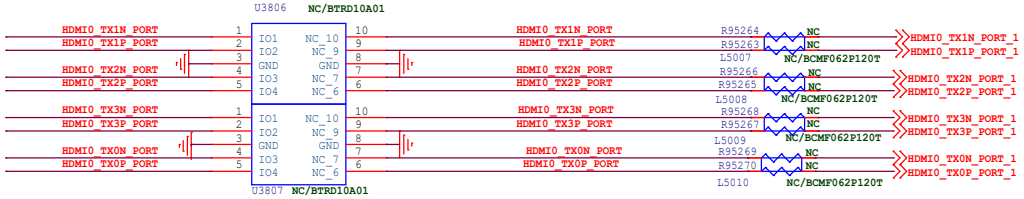




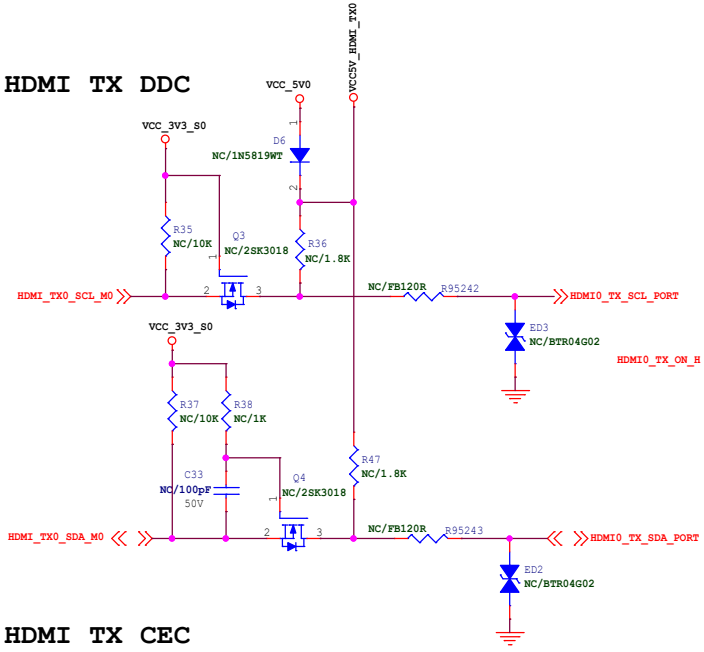
Size: A2	Title: E52C		REV: V1.2
	Page Name: DRAM-LPDDR4/4X_2X32bit		
Date: Thursday, October 10, 2024	Sheet 12 of 22		

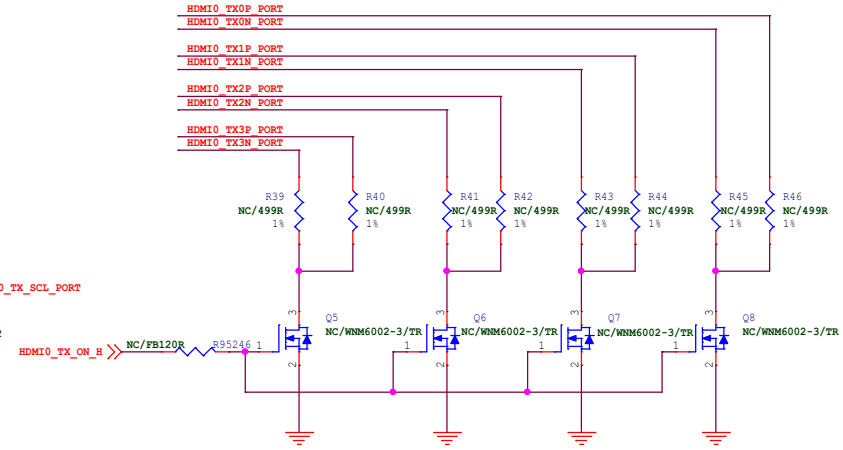
HDMI2.1 TX0

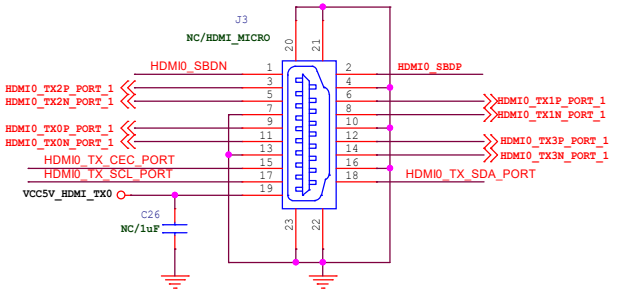




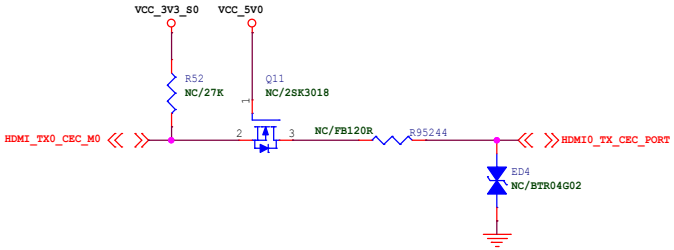
HDMI TX DDC

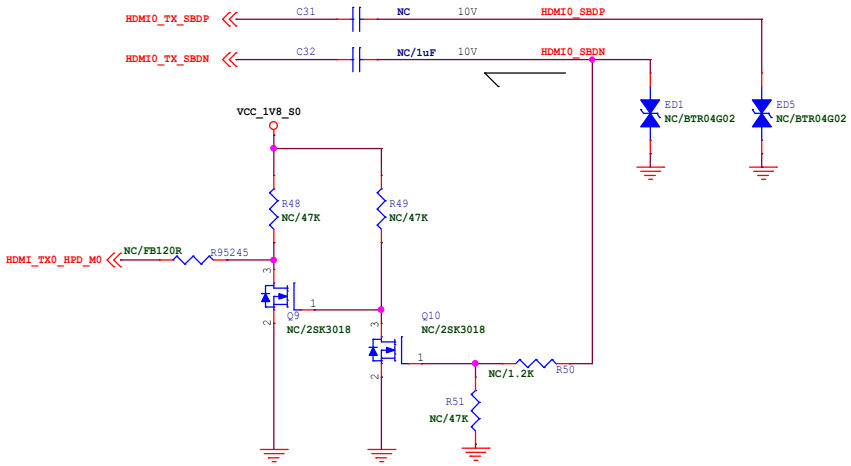




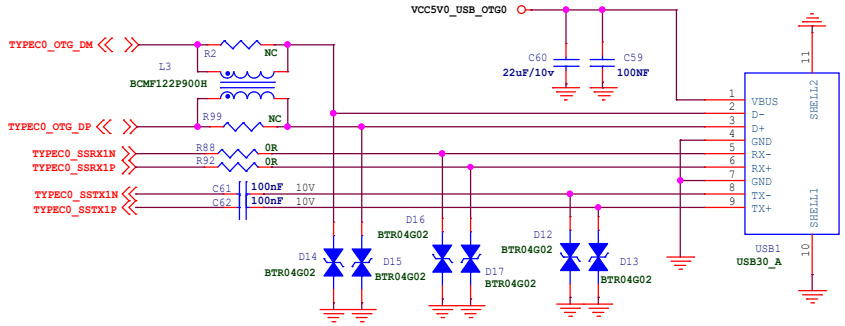


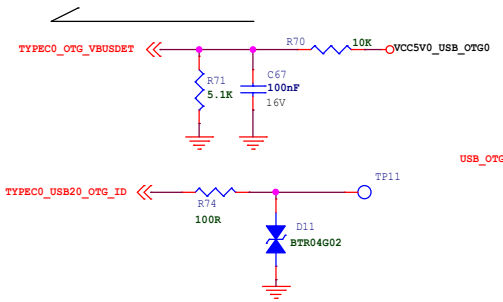
HDMI TX CEC

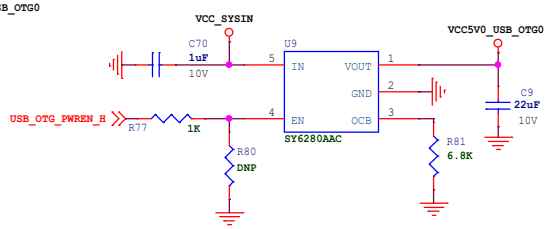


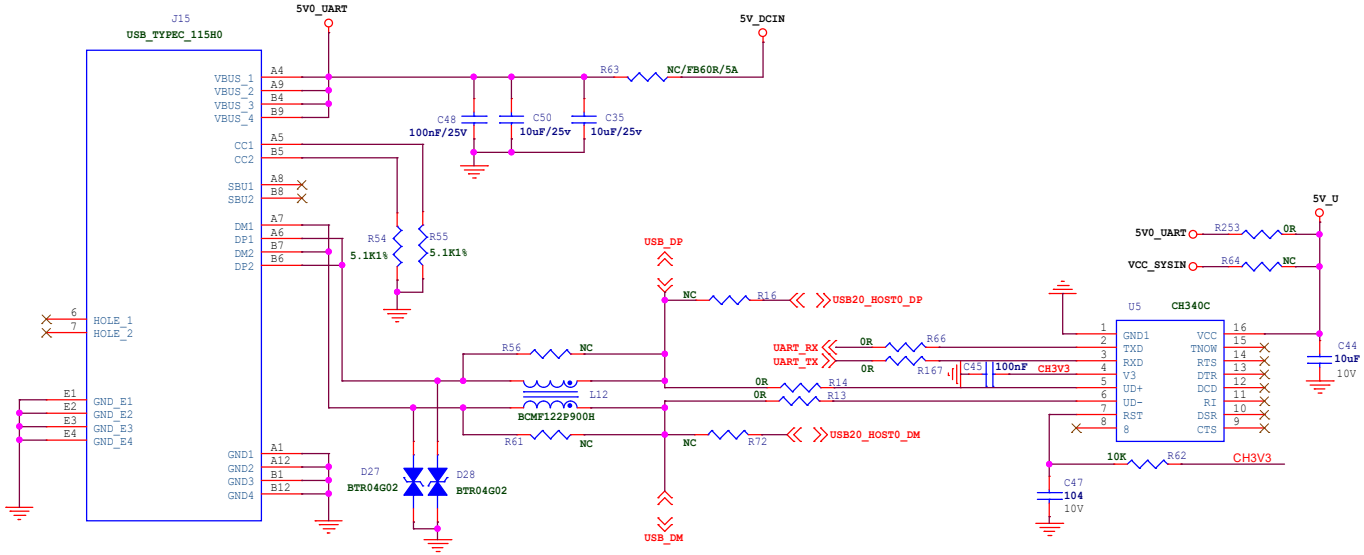


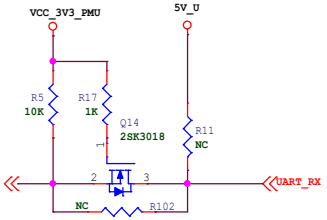
Size A3	Title: E52C		REV V1.2
	Page Name: MICRO HDMI 8K		
Date: Thursday, October 10, 2024	Sheet 13 of 22		

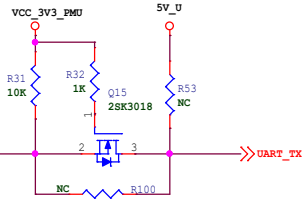






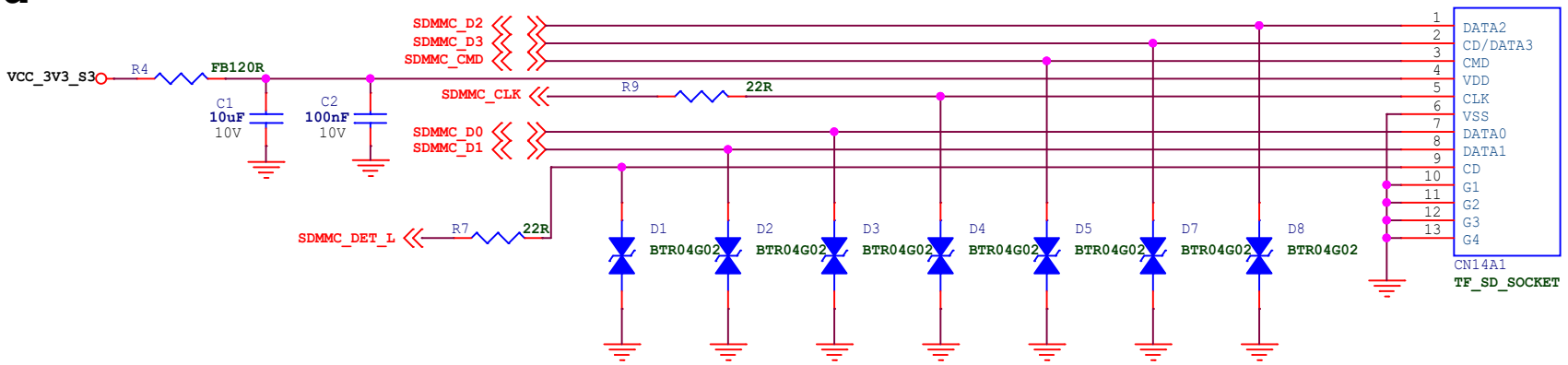




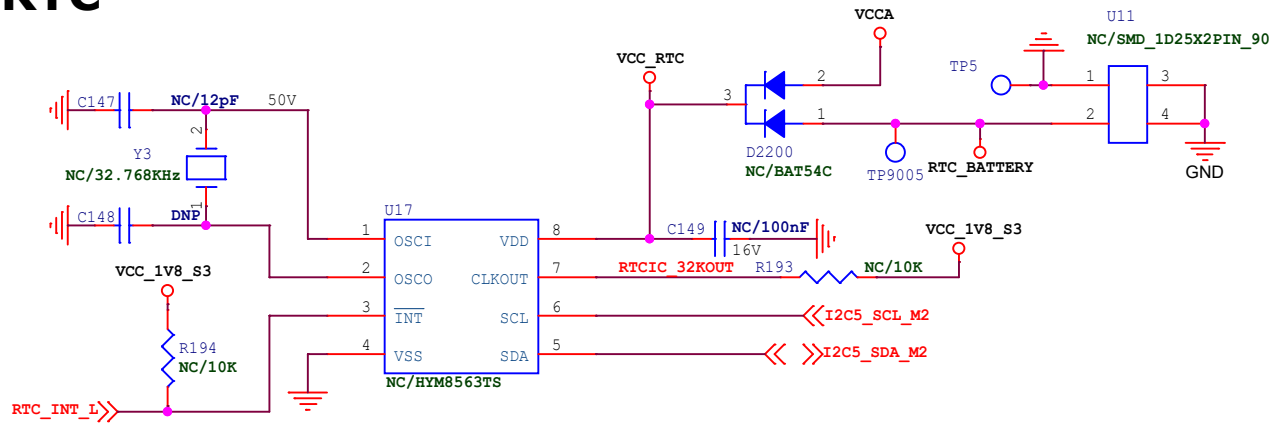


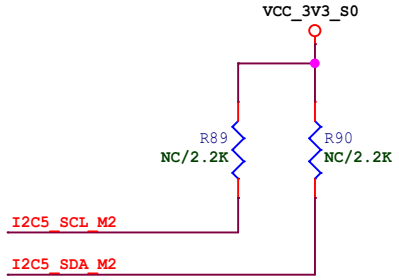
Size A3	Title: E52C		REV V1.2
	Page Name: UART/USB3 Port		
Date: Thursday, October 10, 2024	Sheet 14 of 22		

TF Card

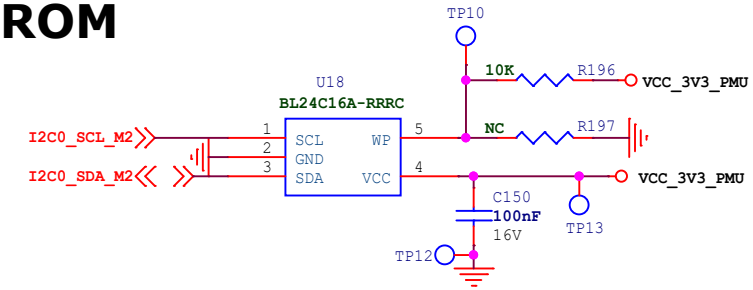


RTC



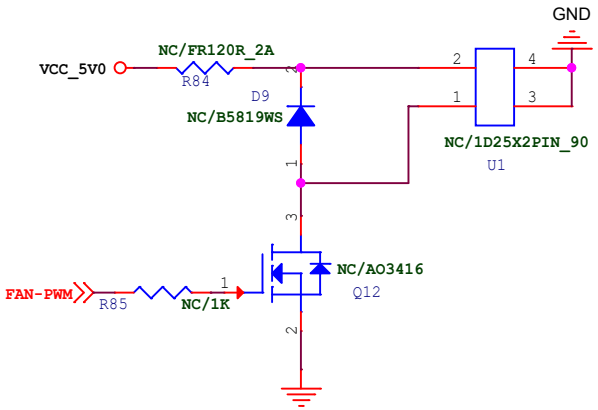


EEPROM

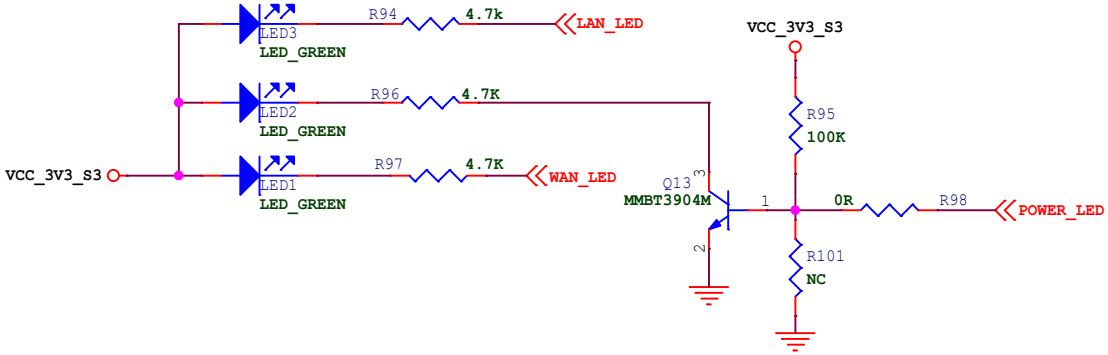


Size A4	Title: E52C		REV V1.2
	Page Name: TF Card/RTC/EEPROM		
Date: Thursday, October 10, 2024		Sheet 15 of 22	

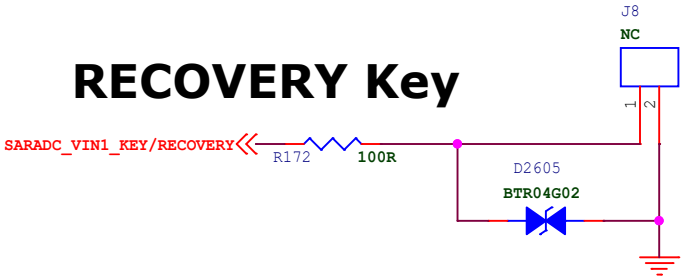
FAN



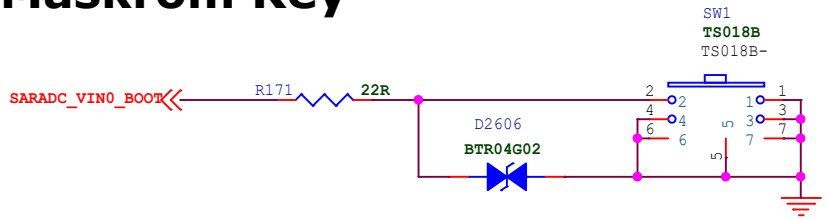
LED

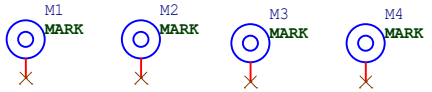


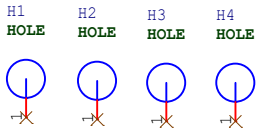
RECOVERY Key



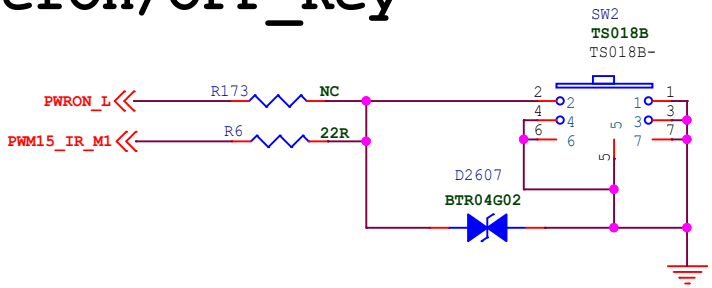
Maskrom Key







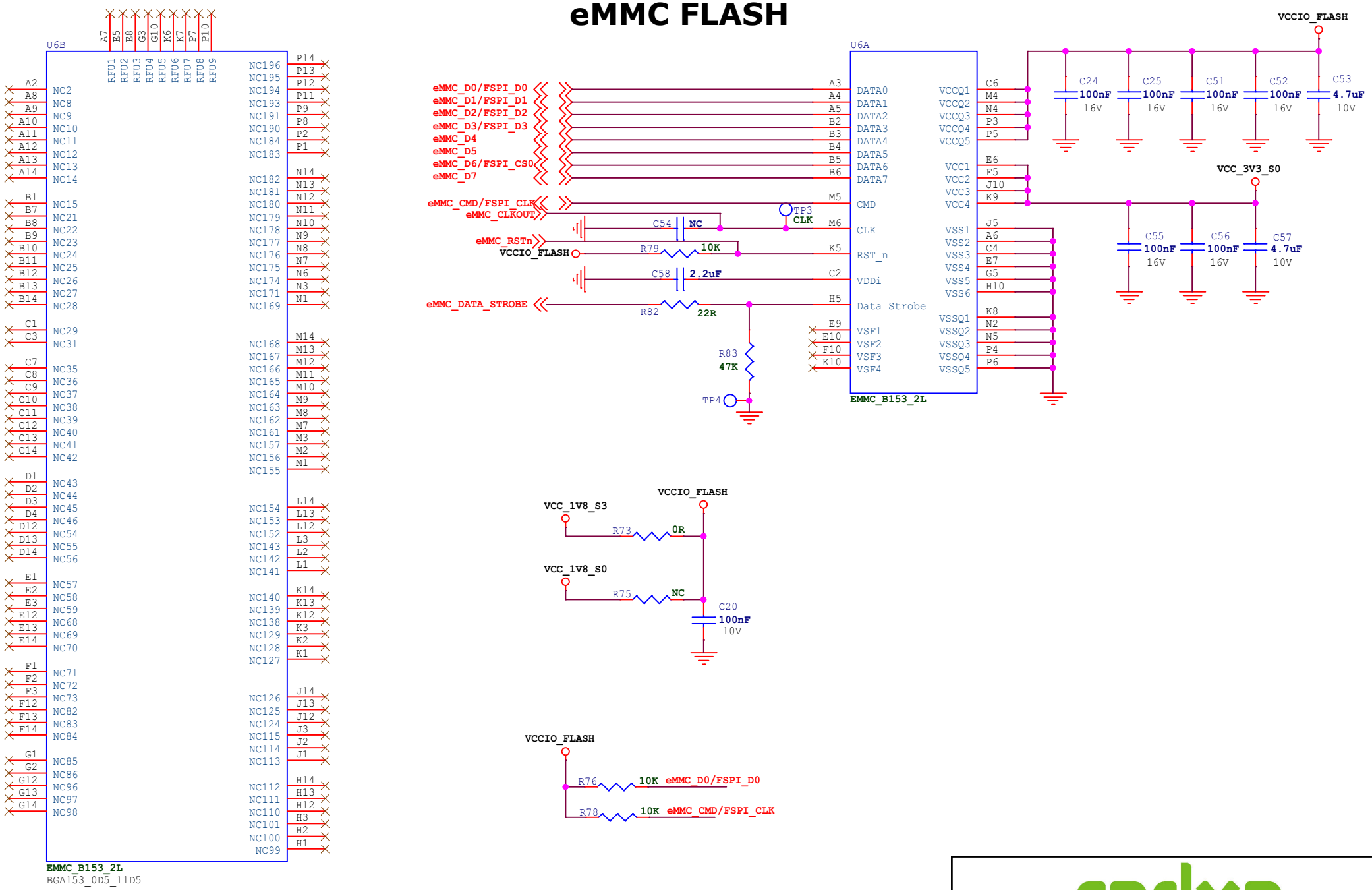
PowerOn/OFF_Key



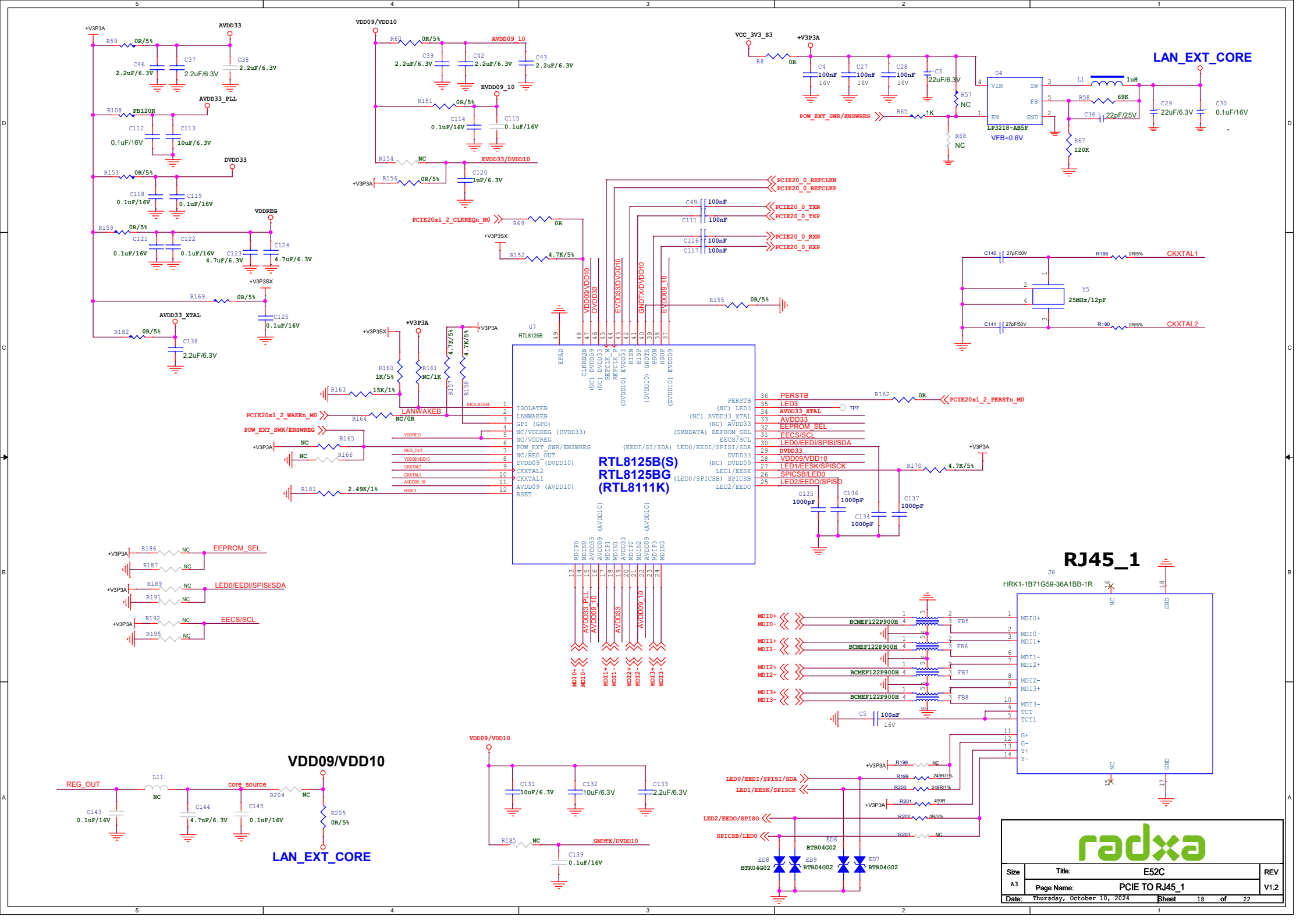
radxa

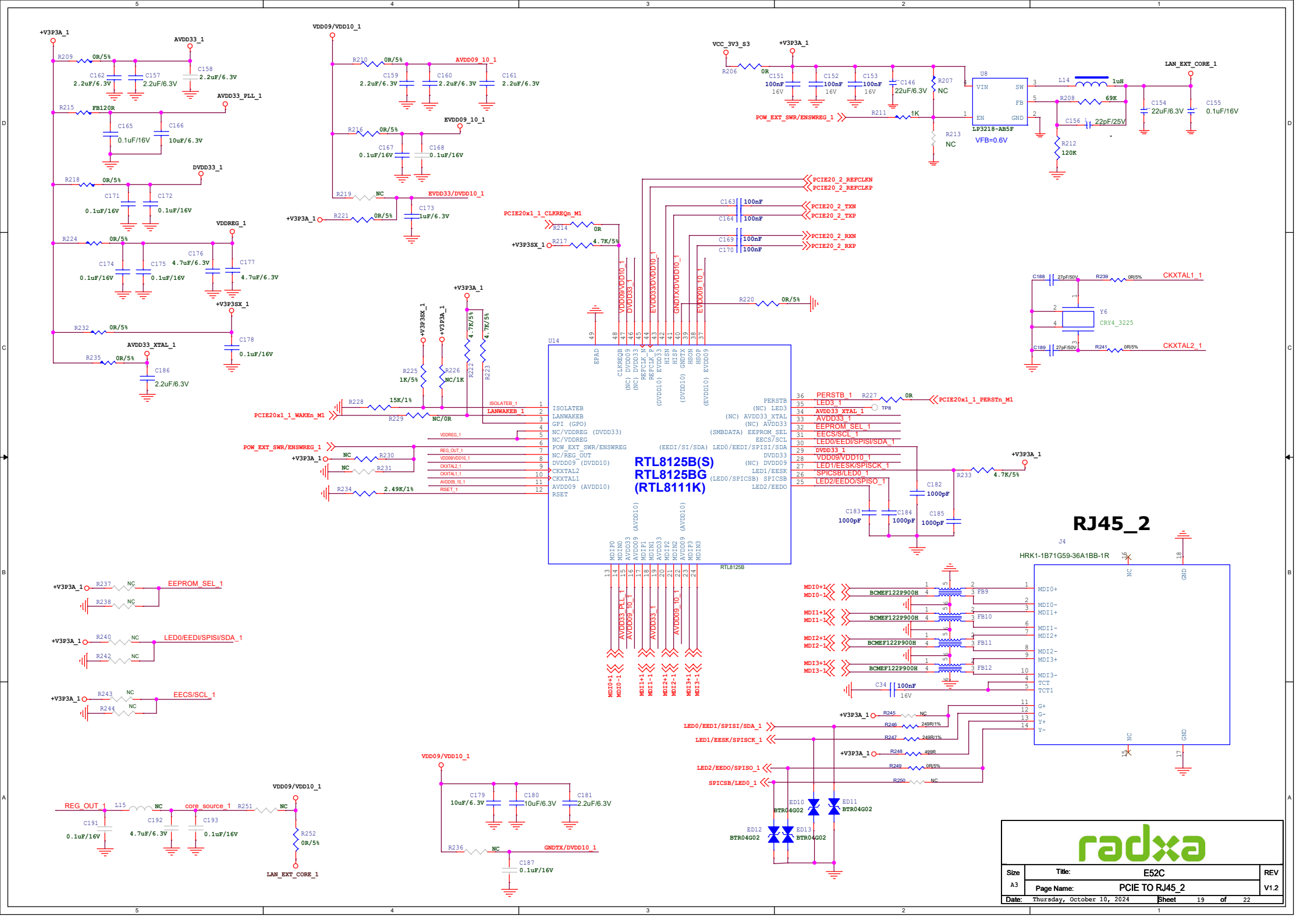
Size	Title:	E52C	REV
A4	Page Name:	KEY/LED/FAN	V1.2
Date:	Thursday, October 10, 2024	Sheet 16 of 22	

eMMC FLASH

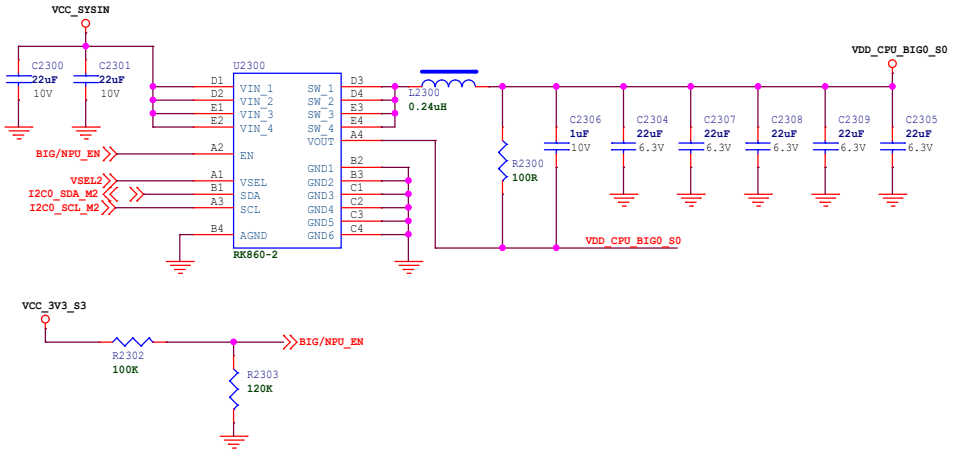


Size A4	Title: E52C	REV V1.2
	Page Name: EMMC FLASH	
Date: Thursday, October 10, 2024	Sheet 17 of 22	

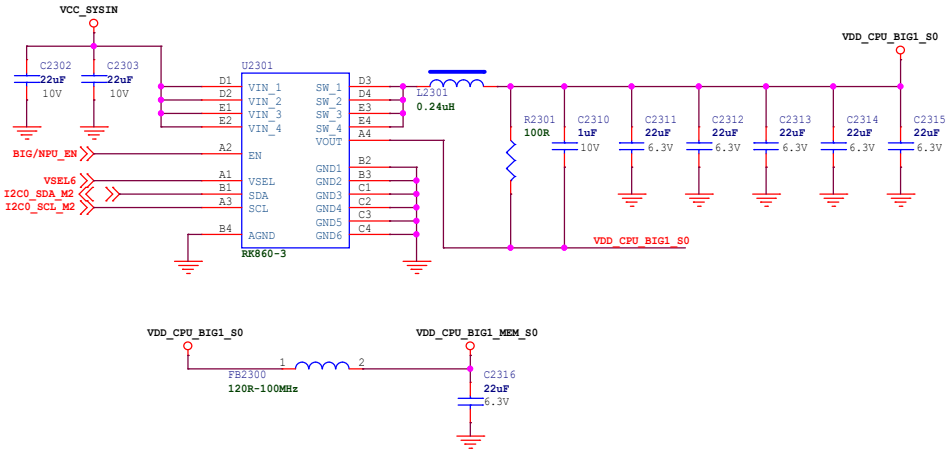

LAN_EXT_CORE
RTL8125B(S)
RTL8125BG
(RTL8111K)
RJ45_1
HRK1-1B71G59-36A1BB-1R
VDD09/VDD10
LAN_EXT_CORE
radxa
Size
A3
Title:
E52C
REV
V1.2
Page Name:
PCIE TO RJ45_1
Date:
Thursday, October 10, 2024
Sheet
18
of
22


RTL8125B(S)
RTL8125BG
(RTL8111K)
RJ45_2
HRK1-1B71G59-36A1BB-1R
LP3218-AB5F
VFB=0.6V
BCMEF122P900H
BTR04G02
Title: E52C
REV
Size A3
Page Name: PCIE TO RJ45_2
V1.2
Date: Thursday, October 10, 2024
Sheet 19 of 22

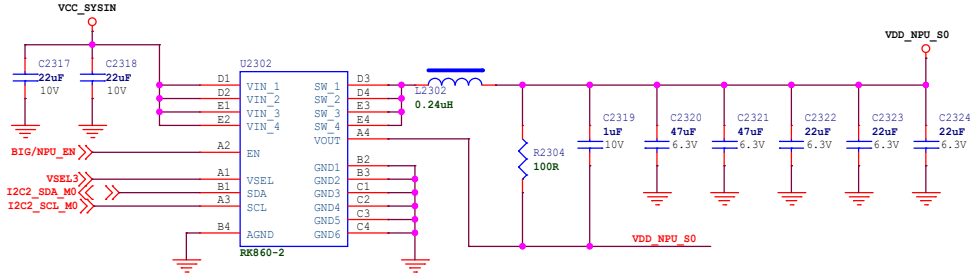
VDD_CPU_BIG0



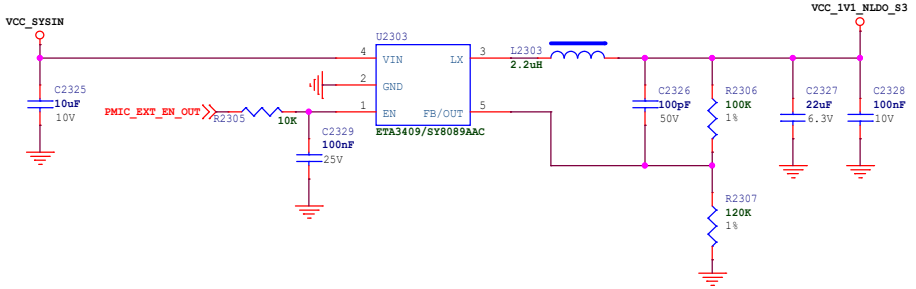
VDD_CPU_BIG1



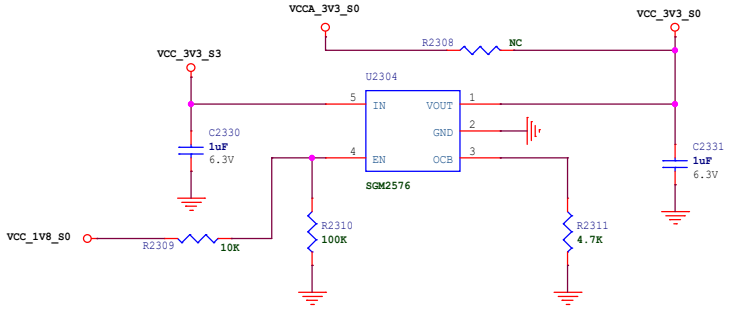
VDD_NPU



VCC_1V1_NLDO



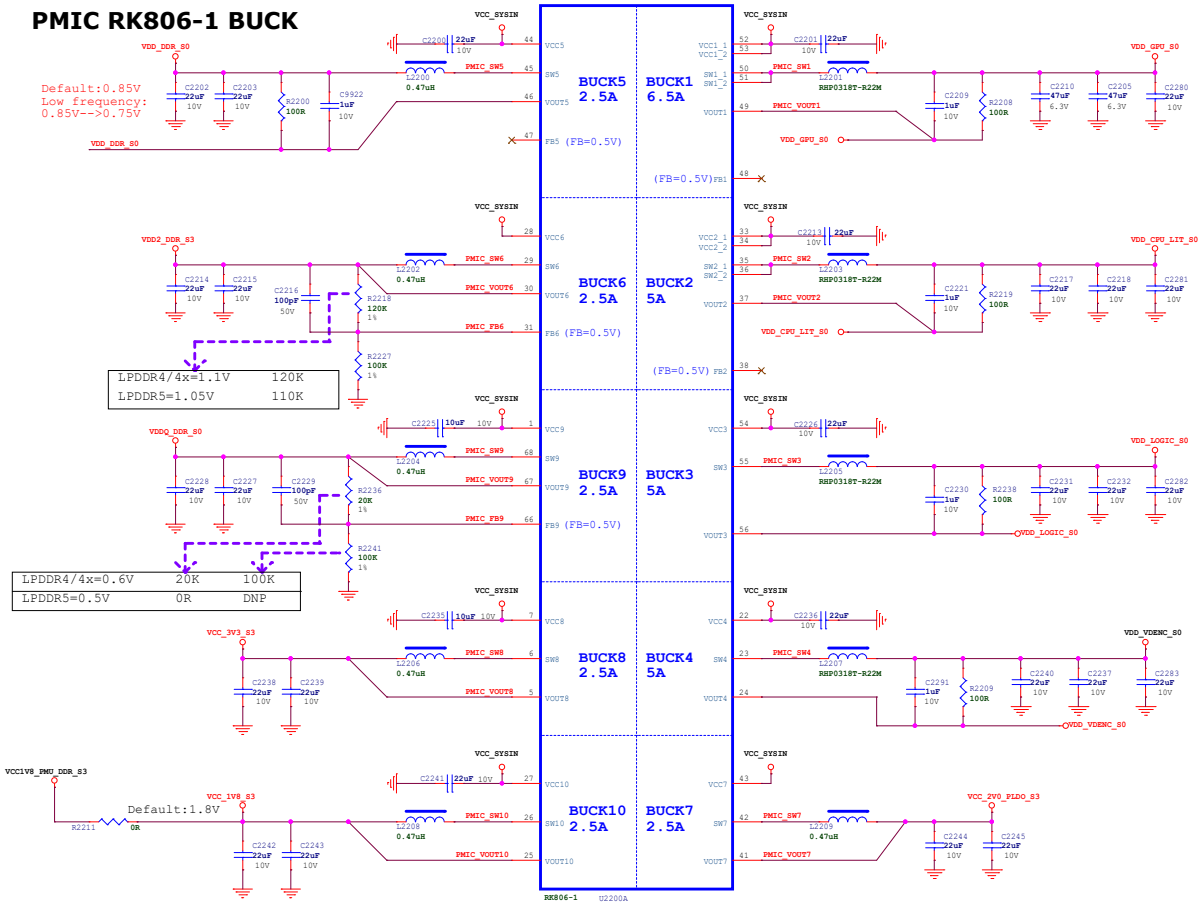
VCC_3V3_S0



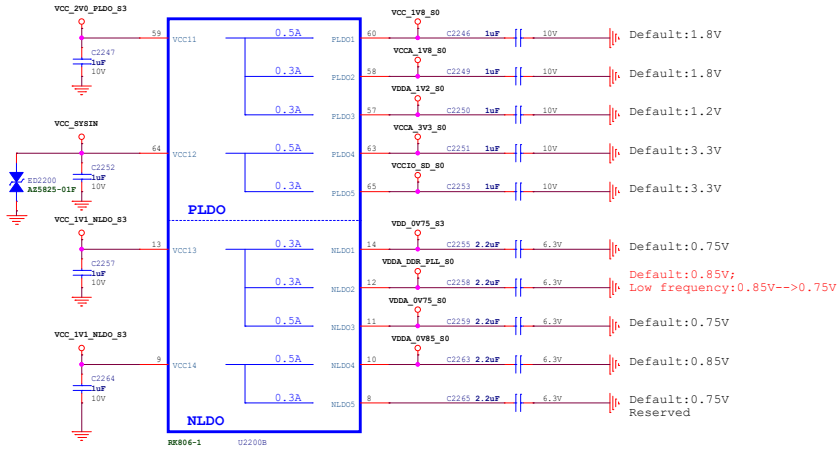
Size A3	Title: E52C		REV V1.2
	Page Name: Power Ext		
Date: Thursday, October 10, 2024	Sheet 20 of 22		

PMIC1 RK806-1

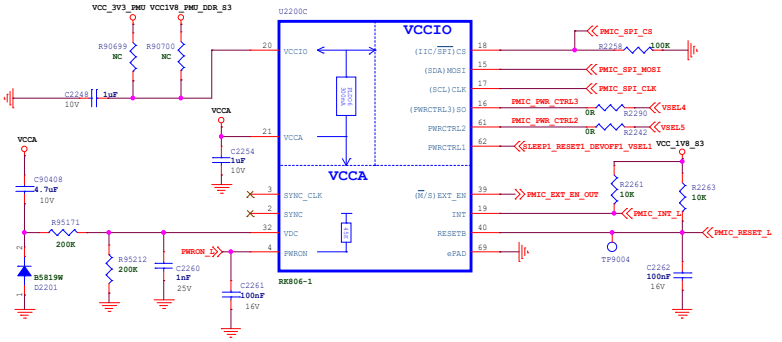
PMIC RK806-1 BUCK



PMIC RK806-1 LDO

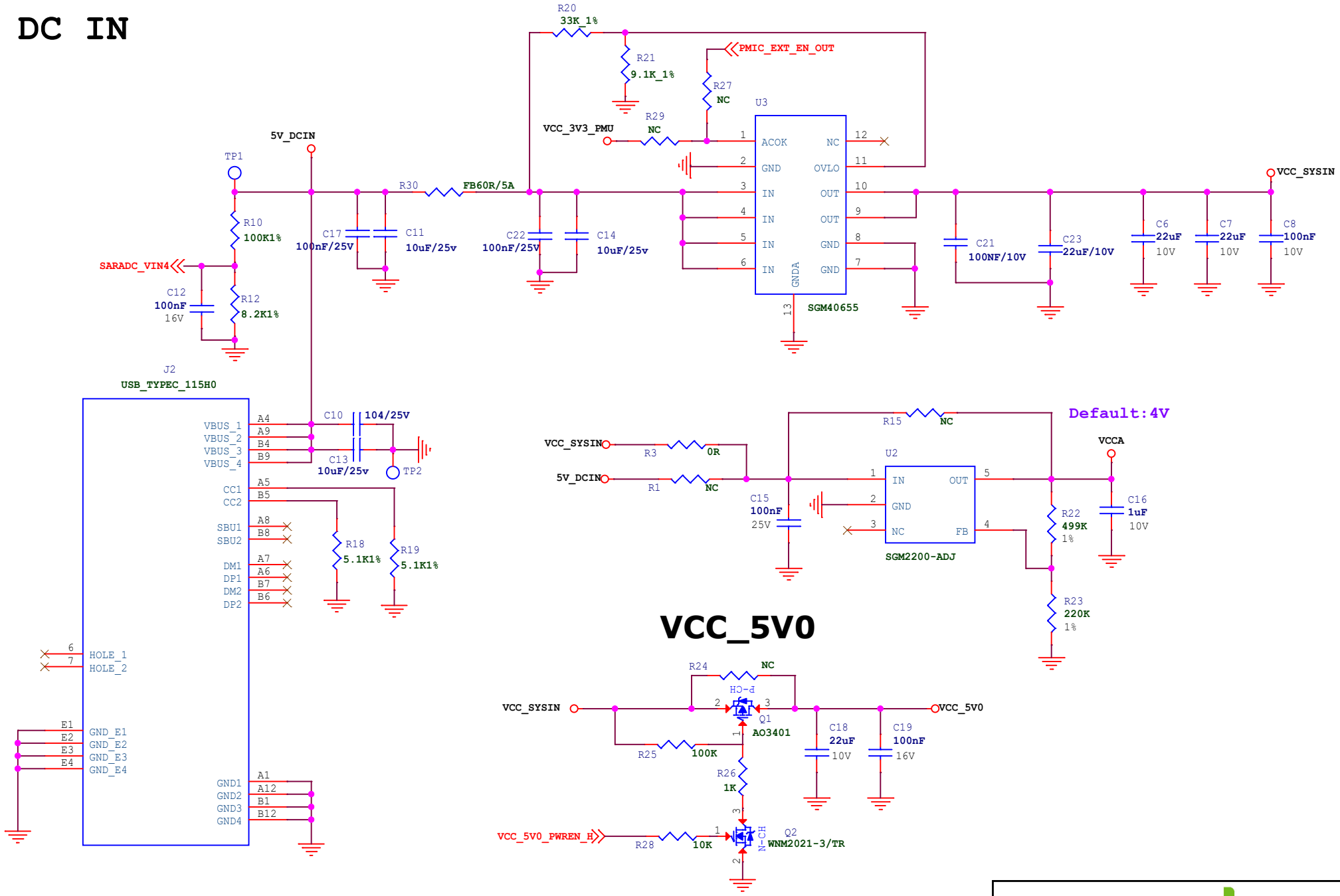


PMIC RK806-1 Managerment



radxa			
Size: A2	Title: E52C		REV: V1.2
	Page Name: Power RK806-1		
Date: Thursday, October 10, 2024	Sheet 21 of 22		

DC IN



Size A4	Title: E52C	REV V1.2
	Page Name: Power DC IN	
Date: Thursday, October 10, 2024	Sheet 22 of 22	